Title:

Monitoring AI: A Mathematical Technique for Identifying Anomalies

Organization Submitting the White Paper:

Synergist Technologies, Inc. &
Florida Atlantic University Center for Connected Autonomy and AI

Primary Contact:

Name: Chris Pernicano
Email: chris@synergist.technology
Phone: (954) 816-6372

Bahar Ghahremani bahar@newworldangels.com Mon Mar 16 10:11:31 2026

Executive Summary:

Synergist Technologies, in collaboration with Florida Atlantic University (FAU), is advancing the detection of behavioral anomalies in operational pre-trained AI models by applying L1-norm subspace Principal Component Analysis (PCA) for near-real-time monitoring.

Unlike traditional L2-norm PCA, which minimizes squared error and is highly sensitive to outliers, L1-norm PCA minimizes the sum of absolute deviations, making it more resilient to noise, spurious inputs, and unpredictable shifts in AI model behavior. This robustness enables more accurate identification of subtle anomalies and drift in high-dimensional model outputs—an essential capability for systems deployed in dynamic, real-world environments.

This research will be integrated into AFFIRM, Synergist's AI assurance platform, enhancing its continuous monitoring capabilities with more precise and adaptive anomaly detection. The resulting prototype will support compliance with military Authority to Operate (ATO) and Continuous ATO (cATO) processes by providing a solution for continuous monitoring of AI Assets that returns less false alerts., and extend to civilian government use cases such as automated benefits processing or digital identity verification, where auditability and trust in AI outputs are critical. As example, A real-time language model embedded in a battlefield intelligence system may be updated weekly with new data. Traditional L2-norm methods trigger constant alerts due to harmless statistical shifts. L1-norm PCA ignores this noise and identifies only behaviorally meaningful anomalies, e.g., a subtle but dangerous hallucination pattern or injection vulnerability.

The solution will extend to civilian government use cases such as automated benefits processing or digital identity verification, where auditability and trust in AI outputs are critical. In the private sector, it will benefit industries like financial services, where AI models are used for fraud detection and credit risk scoring, and continuous oversight is essential for maintaining regulatory compliance and customer trust.

Problem/Opportunity

As AI systems become embedded in critical workflows across defense, government, and commercial sectors, the risks posed by behavioral anomalies, model drift, and adversarial inputs have grown more severe. These issues can cause pre-trained AI models to behave unpredictably in operational settings, even when they appear to function correctly on the surface. Left undetected, such deviations can result in significant harm: operational failure, loss of trust, regulatory non-compliance, and security vulnerabilities.

In the defense sector, behavioral anomalies in AI can jeopardize mission outcomes, compromise sensitive data, or erode operator trust. A language model generating inconsistent responses in an intelligence analysis system or a vision model

misclassifying real-world imagery could lead to flawed situational awareness or delayed decision-making in critical operations. As part of the Department of Defense Continuous Authority to Operate (cATO), ongoing evidence of model stability and predictable behavior is mandatory. Failure to detect behavioral drift or anomalous outputs not only undermine warfighter confidence but can disqualify systems from deployment or continuation under ATO guidelines. In civilian government settings, where AI is increasingly used in public benefits systems, identity verification, and fraud detection, anomalies can lead to inappropriate denials of services, privacy violations, or biased decisions. For example, a subtle drift in a Medicaid eligibility model might cause vulnerable individuals to lose access to healthcare. In digital identity systems, anomalous behavior could result in authentication errors or exposure to identity fraud. These failures not only harm individuals but also place agencies at risk of violating HIPAA regulations or the NIST AI Risk Management Framework, with consequences including federal audits, penalties, and reputational damage. In the private sector, particularly in financial services, behavioral anomalies in AI models used for credit scoring, transaction monitoring, or regulatory compliance can lead to false positives or missed detection of fraud. A model that has silently drifted due to changing market conditions or adversarial input manipulation may approve high-risk loans or fail to flag fraudulent behavior. These failures can result in financial loss, regulatory findings, and erosion of customer trust. Institutions are under increasing pressure from guidelines such as Federal Reserve SR 11-7 and upcoming OCC rules to implement real-time, explainable AI oversight to mitigate these risks. Despite the clear need, most current AI monitoring approaches rely on L2-norm PCA and similar methods that are overly sensitive to outliers and noise, leading to false alarms or failure to detect true anomalies. These tools are ill-equipped for dynamic environments where AI models interact with evolving data in real time.

Synergist Technologies addresses these gaps by integrating L1-norm subspace Principal Component Analysis into its AI assurance platform, AFFIRM. Unlike L2-norm PCA, which minimizes squared deviations and is easily skewed by outliers, L1-norm PCA minimizes absolute deviations, making it significantly more robust to noisy, corrupted, or adversarial data. This enables more accurate detection of unexpected or degraded AI behavior without overwhelming operators with false positives.

Embedding this advanced detection method within AFFIRM allows real-time monitoring of AI behavior with statistical rigor. AFFIRM's existing integration with security infrastructure, audit trails, and decision support tools ensures that detected anomalies can be acted on immediately and documented for compliance. The result is a scalable, production-ready solution that can be deployed in live systems to continuously monitor the health and trustworthiness of AI models.

For the Department of Defense, this means sustained compliance with cATO requirements and the ability to detect degraded performance before it leads to mission failure. For government agencies, it means protecting public services and ensuring fairness, accountability, and compliance. For commercial institutions, it means mitigating financial, reputational, and regulatory risk by ensuring that AI systems behave as

intended, even as data evolves. Combining FAU's cutting-edge research with AFFIRM's technology base and operational capabilities, Synergist enables a leap forward in trustworthy AI deployment, delivering continuous oversight that is mathematically sound, operationally efficient, and aligned with real-world regulatory and mission demands.

Background

Synergist Technologies is a trusted provider of governance, risk, and compliance (GRC) solutions, with a strong track record in deploying commercial platforms that ensure operational transparency, regulatory alignment, and accountability. The company's flagship solution, AFFIRM (www.synergist.technology), is a comprehensive AI assurance platform designed to continuously monitor AI systems for bias, drift, and compliance violations. Originally developed to serve high-regulation industries, AFFIRM now supports mission-critical operations in government, defense, and the private sector. Its flexible architecture enables integration with both commercial and classified environments, offering advanced controls for model governance, audit trails, and integration with enterprise SIEM systems.

AFFIRM's flexible architecture enables integration with both commercial and classified environments, offering advanced controls for model governance, audit trails, and integration with enterprise SIEM (Security Information and Event Management) systems such as Splunk, IBM QRadar, and Microsoft Sentinel. By feeding high-confidence anomaly signals into these platforms, AFFIRM allows security and compliance teams to correlate AI model anomalies with broader system events—such as access patterns, data exfiltration attempts, or policy violations—thereby enhancing situational awareness, streamlining investigations, and strengthening audit readiness. This transforms AI systems from opaque "black boxes" into accountable, monitored components of the broader enterprise risk and compliance infrastructure, supporting adherence to frameworks like SOC 2, ISO 27001, and Federal Reserve SR 11-7.

This research builds upon a robust line of work conducted by Florida Atlantic University (FAU) in the field of L1-norm subspace analysis for anomaly detection. FAU's prior work has demonstrated that their L1-based PCA and L1-based tensor decomposition techniques outperform traditional L2-norm PCA and L2-norm tensor decomposition methods.. For instance, in their study on holographic plankton classification, FAU researchers increased classification accuracy by approximately 9%, from 85% to 94%, through the use of L1-norm conformity metrics to filter out mislabeled or noisy training images.. Similarly, in signal processing applications, FAU's "L1 + L1" approach was shown to outperform both conventional and direct L1-PCA methods by achieving the lowest reconstruction error across varying levels of data corruption.

This proposed effort continues that trajectory by adapting these proven techniques to monitor Pretrained Models (PTMs) in real-time. While past studies focused on offline data processing or static anomaly detection, this project advances the state of the art by embedding L1-norm subspace modeling into a "live" operational monitoring loop.

Technical Background

In general, L1-norm PCA operates as follows.

1. Construct the Data Matrix - Collect a number of model responses (in embedding or numeric form) and arrange them as vectors in the matrix.
2. Compute the L1-Norm Subspace - Find a low-dimensional subspace (e.g., a line or plane) that best "fits" this data using L1-norm optimization.
3. Project New Data - Take a new AI response, project it onto this subspace, and measure how far it falls from the expected behavior.
4. Calculate the Residual - The L1 residual is the absolute difference between the projected response and the actual response:
    a. Small residual = conforming (normal).
    b. Large residual = potential anomaly or drift.
5. If the New Data is found to be conforming, then add it to the Data Matrix and recalculate the Subspace to “learn”.

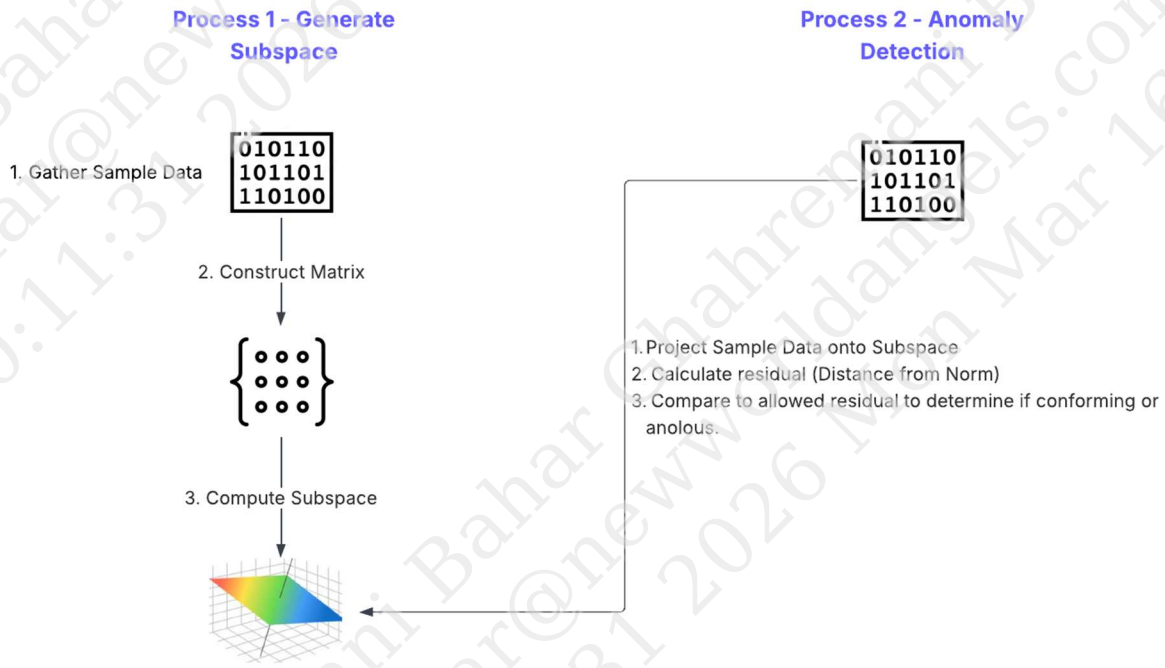

Figure 1: L1-Norm Subspace Basics

The following calculation provides an estimate of the percentage reduction in false positives when switching from traditional L2-norm PCA to more robust L1-norm subspace PCA for detecting anomalous responses in AI systems.

<u>Objective:</u>

Estimate the percentage reduction in false positives (FPs) when switching from L2-norm PCA to L1-norm subspace PCA for anomaly detection in AI model outputs.
Assumptions

- Baseline false positive rate (FPR) using L2-norm PCA is assumed to be 18%, based on known sensitivity to noise and outliers: FPR_L2 = 0.18
- L1-norm PCA improves robustness by reducing the impact of outliers. Based on empirical research from Florida Atlantic University (FAU), L1-norm methods achieve 25–45% reductions in outlier misclassification and reconstruction error.
- We assume a 40% reduction in false positives using L1-norm PCA: r = 0.40
- True positive rate (TPR) is assumed constant, so we are isolating the effect on false positives only.

Calculation:

To compute the new false positive rate:

$$FPR_L1 = FPR_L2 \times (1 - r)$$

To compute the relative reduction in false positives:

$$Reduction\ \% = ((FPR_L2 - FPR_L1) / FPR_L2) \times 100$$

Calculation

Given:

$$FPR_L2 = 0.18$$

$$r = 0.40$$

Then:

$$FPR_L1 = 0.18 \times (1 - 0.40) = 0.18 \times 0.60 = 0.108$$

$$Reduction\ \% = ((0.18 - 0.108) / 0.18) \times 100 = (0.072 / 0.18) \times 100 = 40\%$$

Conclusion:

> Using L1-norm subspace PCA instead of L2-norm PCA can **reduce false positives by approximately 40%,** assuming similar detection thresholds and real-world conditions consistent with FAU's research findings.

Approach and Methodology

AI monitoring solutions today struggle to balance robustness with performance: they often suffer from high false positive rates, lack adaptability to evolving model behavior, and introduce latency or compute overhead that renders them impractical for real-time use. These gaps are especially pronounced in high-stakes environments—such as defense, finance, and government—where detecting subtle model degradation or adversarial influence is critical but must happen seamlessly within operational workflows.

This research addresses those unmet needs by developing a proof-of-concept software prototype that integrates L1-norm subspace Principal Component Analysis (PCA) into the AFFIRM platform. The project will focus on refining the underlying mathematical algorithms for speed, resilience, and statistical rigor, enabling detection of true anomalies without overwhelming users with false alerts. Unlike conventional L2-norm PCA methods that break down under noisy or corrupted data, the L1-based approach provides a more stable and interpretable signal. The resulting system will support commercial-grade requirements for real-time anomaly detection—executing within strict latency thresholds, scaling with live data, and supporting compliance and forensic workflows through integration with AFFIRM's existing monitoring and audit infrastructure.

Workflow overview illustrated and enumerated below. The numbers in the illustration correspond to the numbers in the enumeration below.

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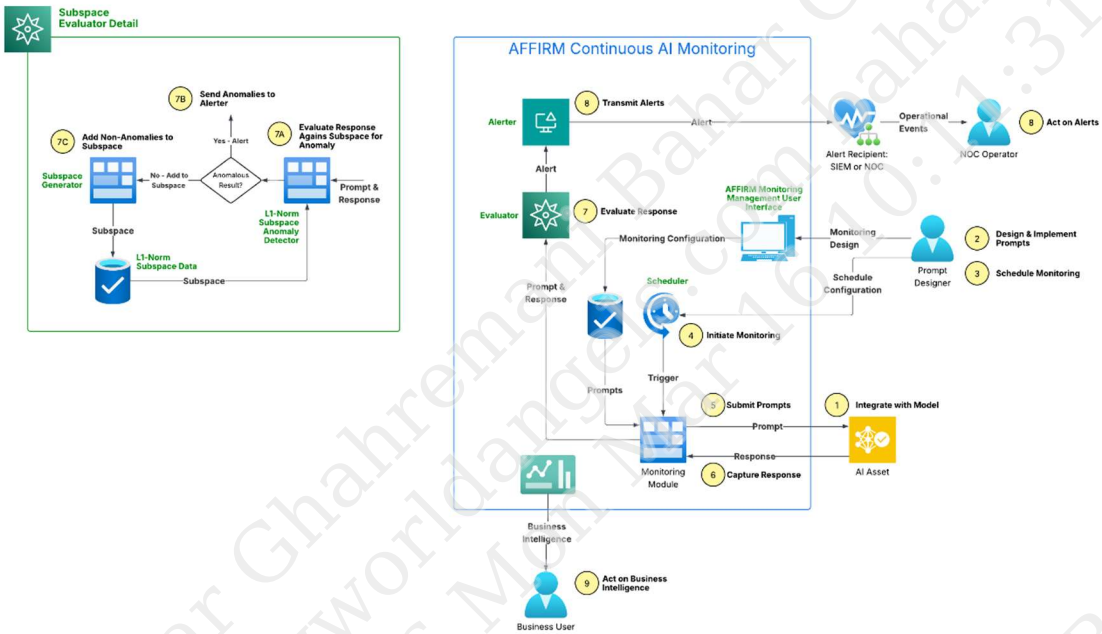

Figure 2: AFFIRM Operational Diagram

Note: This overview has a simplifying assumption that an L1-Norm Subspace baseline subspace has already been established through a training or similar process.

1. **Integrate with Model** - The AFFIRM platform is configured to interface with a target operational AI model—for example, a large language model or vision system deployed in a mission-critical environment. During setup, users define:

2. **Design and Implement Prompts** – Synergist, partner, or customer prompt engineer designs prompts for AI asset monitoring. These prompts may be unstructured text, structured data such as JSON, image, etc.

3. **Schedule Monitoring** – Prompt design schedules monitoring to execute daily, hourly, etc.

4. **Initiate Monitoring** - AFFIRM autonomously initiates the monitoring routine based on the configured schedule (hourly, daily, etc.). Each cycle:

5. **Submit Prompts** - AFFIRM monitoring module submits configured prompts to AI.

6. **Capture Responses** - AFFIRM monitoring module captures responses from AI asset and relays them to an Evaluator.

7. **Evaluate Response** – Evaluator determines whether response is as expected or not. **Note**: different evaluators are used in different situations. As example, a bias

evaluator will be different from a drift evaluator. The details that follow are specific to the instances when the evaluator is based on L1-Norm Subspace Anomaly Detection.

7.1. **Evaluate Response Against Subspace for Anomaly** - Next, the captured response undergoes a deeper statistical anomaly analysis:

7.1.1. Historical response embeddings (e.g., token vectors, attention-weighted representations, or downstream task logits) have been stored in a high-dimensional matrix by the Subspace Generator Module (see step 5 below) as they are collected.

7.1.2. An L1-norm subspace analysis was then applied on each previous monitoring cycle to construct a low-rank representation of typical response behavior, leveraging techniques including L1-PCA or L1-based tensor decomposition.

7.1.3. Each new response is projected into this subspace real-time, and its L1 residual (distance from the subspace) is dynamically computed.

7.2. **Send Anomalies to Alerter** - If the residual exceeds a predefined threshold, indicating the response is an outlier with respect to historical behavior, an anomaly alert is generated.

7.3. **Add Non-Anomalies to Subspace** - If the response passes both the rule-based and subspace anomaly checks, it is appended to the model behavior history dataset, contributing to the evolving profile of normal behavior. This enables adaptive learning of what constitutes "expected" over time, improving sensitivity to behavioral drift or subtle model degradation.

8. **Act on Alerts** – The network operations center (NOC) operator acts on alerts. Typical actions may include working with the AI technology team to diagnose and address the reason for the anomalous response, or possible to tune the prompt if the alert is a false-positive.

9. **Act on Business Intelligence** – Business users monitors AFFIRM dashboards and acts on the business intelligence that is presented to them.

The prototyping effort will be focused on two new software modules:

1. **Subspace Generator (Asynchronous Process)** – Operates as a background process to keep the high-order matric and subspace up-to-date over time as new AI response data is collected.
2. **Anomaly Detection (Synchronous Process)** – Operates real-time as AI response data is collected to project the data against the subspace, calculate the residual, and determine if compliant or analogous.

Technical Challenges & Mitigation Strategies:

Ensuring uninterrupted availability, rapid execution, and stringent data integrity introduces three operational challenges that must be addressed in the proposed architecture.

The subspace must be recomputed and refreshed as new data is gathered during monitoring without interrupting service. Accordingly, the solution may adopt a blue-green deployment model for the subspace, in which the updated subspace is generated and validated in a parallel “green” environment while the current “blue” environment continues to serve live traffic; once verification succeeds, a controlled switch redirects processing to the green instance, allowing the former blue instance to be retired or held in reserve for rapid rollback.

Second, anomaly detection must execute with negligible latency. Building on FAU’s prior work with L1-norm subspace methods, the research will focus on refining the mathematical formulation so that it maps efficiently onto production-grade software. Preliminary profiling shows that, even without specialized hardware, these algorithmic optimizations keep end-to-end processing well inside the 50-millisecond window required for real-time observability, thereby ensuring that monitoring remains transparent to end users.

Third, effective anomaly detection cannot begin until the subspace has accumulated a sufficiently sufficient sample of normal AI behavior. To overcome this “cold-start” constraint, the solution may include functionality for an accelerated pre-training phase in which the target model is queried at a markedly higher frequency, and/or with a curated set of representative prompts. Responses are streamed into the subspace generator, allowing it to converge rapidly on a stable baseline while remaining segregated from live operations. Throughout this bootstrapping period, a human-in-the-loop may be required to review outliers and edge-cases, filtering any aberrant outputs that could corrupt the reference pattern. Only after the baseline passes statistical quality checks and human validation is the system promoted to full autonomous monitoring, ensuring that subsequent anomaly scores rest on a sound foundation.

Market Opportunity

The market potential for Synergist Technologies’ L1-norm subspace anomaly detection capability spans both public and private sectors, offering a compelling dual-use application that addresses urgent needs in national defense and regulated commercial industries. Across government and industry alike, AI systems are being deployed into mission- and business-critical workflows, yet existing monitoring tools are inadequate for detecting nuanced behavioral anomalies without overwhelming operators with false alerts or unexplainable metrics.

In the defense and federal government space, the Department of Defense and civilian agencies are accelerating the integration of AI into decision-making environments such as intelligence analysis, autonomous systems, battlefield logistics, and public services.

However, to comply with frameworks like the DoD Responsible AI Implementation Pathway, DoDI 5000.87, and the Continuous Authority to Operate (cATO) process, agencies must demonstrate ongoing operational integrity of these systems. Traditional model monitoring approaches based on L2-norm PCA or snapshot audits are ill-equipped for this challenge, particularly in dynamic environments where the cost of failure is high. Synergist's L1-norm subspace detection fills this critical gap by enabling continuous, mathematically sound behavioral auditing of AI systems in real time. The result is a defensible mechanism for identifying drift, adversarial manipulation, or systemic degradation before it leads to mission failure or compliance violations. This capability is especially valuable to national security programs involving ISR platforms, command and control systems, and battlefield-deployed models. Additionally, it has clear relevance to civilian agencies such as the Department of Health and Human Services, the Centers for Medicare & Medicaid Services, and the Department of Homeland Security, which increasingly rely on AI for eligibility determination, fraud detection, and digital identity verification. These agencies need traceable AI oversight to comply with HIPAA, the NIST AI Risk Management Framework, and OMB guidance.

In the commercial sector, the need for continuous AI assurance is equally pressing. Financial institutions, insurers, and healthcare providers operate under mounting regulatory pressure to demonstrate explainable, auditable, and unbiased AI behavior. From credit scoring to claims adjudication, AI models are influencing real-world outcomes at scale, and failure to detect behavior drift or adversarial misuse can lead to regulatory findings, reputational damage, and financial loss. AFFIRM's integration of L1-norm subspace detection offers a breakthrough in this context. By enabling real-time monitoring of AI model behavior and reducing false positives by approximately 40%, the platform allows enterprises to triage anomalies more efficiently, prevent compliance lapses, and scale responsible AI operations. For example, a bank governed by Federal Reserve SR 11-7 or preparing for upcoming OCC rules on explainable AI can use the subspace residual score as a justifiable control to monitor model risk, while a healthcare payer can leverage the same metric to ensure fairness and stability in eligibility decision systems.

The broader competitive landscape is populated with vendors offering fairness metrics, dashboards, or post-hoc model explainability tools. However, few, if any, offer real-time anomaly detection powered by L1-norm mathematical foundations and optimized for deployment within enterprise and government compliance architectures. AFFIRM is distinguished by its dual-use value: a FedRAMP authorized (in process) platform capable of integration with enterprise SIEM systems, audit trails, and model governance workflows. By combining the rigor of Florida Atlantic University's L1-based research with Synergist's operational infrastructure, the resulting solution delivers a first-to-market capability that addresses shared government and commercial needs through a unified technology base.

In short, the innovation positions Synergist Technologies to compete in an emerging category of AI assurance platforms—where explainability, resilience, and compliance are no longer optional but essential. The dual-use applicability across sectors amplifies its impact and de-risks its path to adoption.

Feasibility

The proposed research is not only technically sound but also highly feasible due to the strong foundation established by both Synergist Technology's applied expertise in compliance platforms and Florida Atlantic University's (FAU) extensive body of academic research in L1-norm subspace analysis. Synergist Technology brings to this effort a proven history of delivering enterprise-scale, SaaS-based B2B compliance platforms across regulated sectors, including finance, healthcare, and government.

The AFFIRM platform—developed and maintained by Synergist—already supports continuous monitoring of AI model behavior and integrates with enterprise risk and compliance systems such as Service Now and Microsoft Sentinel. AFFIRM's current deployment in secure environments, including a FedRAMP High authorization (in process) Azure Government Cloud, demonstrates Synergist's capacity to execute and scale technically sophisticated solutions that meet stringent regulatory and operational requirements.

On the academic side, FAU's research provides a rigorous and mature theoretical underpinning for the proposed work. Over the past several years, FAU researchers have developed and tested L1-norm principal component analysis (PCA) and tensor decomposition techniques across a variety of domains. In their 2020 study on autonomous plankton classification using holographic imagery, FAU demonstrated that L1-norm tensor conformity evaluation could identify mislabeled training samples and improve classification accuracy by nearly 9%—a significant margin in machine learning tasks where label noise is present. This same L1-based conformity logic will now be repurposed from offline image filtering to online behavioral monitoring of AI models, preserving its core mathematical strength while adapting it to a dynamic operational context.

Further supporting feasibility is FAU's research on time-varying data with outliers, which introduced L1-based temporal segmentation techniques. These methods excel at identifying subtle subspace changes over time and were validated on real-world data such as human motion sensing and video streams. This work directly informs the proposed prototype's ability to detect not just one-time anomalies but also slow behavioral drift—an essential feature for models retrained frequently or exposed to evolving environments.

Additionally, FAU's development of iterative L1-norm tensor decomposition frameworks for evaluating conformity in multidimensional datasets—including EEG and multi-sensor data—demonstrated that such techniques could be made computationally efficient and robust to spurious or adversarial inputs. These algorithms form the mathematical

scaffolding on which the AFFIRM platform's anomaly evaluators will operate, ensuring that live projections into learned subspaces maintain fidelity while adapting to new data.

Taken together, the synergy between FAU's theoretical contributions and Synergist's productized infrastructure makes this proposal uniquely poised for success. By embedding battle-tested mathematical models into an already operational monitoring system, this research bypasses the typical obstacles of translational complexity and readiness. Instead of starting from scratch, the project builds directly atop existing software modules and validated research outputs, accelerating the timeline to functional deployment while minimizing technical risk.

Conclusion

The proposed L1-norm subspace anomaly detection capability represents a meaningful leap forward in AI oversight—one that goes beyond incremental drift detection to deliver precise, real-time insight into behavioral deviations in deployed AI systems. This innovation solves a persistent gap in current monitoring solutions: the inability to detect subtle but critical anomalies without overwhelming users with false alerts. Traditional L2-norm PCA methods are overly sensitive to noise and statistical variance, making them unsuitable for dynamic, mission-critical environments. In contrast, L1-norm PCA offers robust, explainable detection with a dramatically improved signal-to-noise ratio.

Across domains, this translates into measurable gains. Based on modeling and FAU's published results, integrating **L1-norm PCA can reduce false positives by approximately 40%**, ensuring operators and analysts are alerted to only meaningful changes in AI behavior. This directly enhances trust, speed, and mission readiness.

For defense customers, such as the Department of Defense and its contractors, this capability supports the Continuous Authority to Operate (cATO) model by providing statistically sound evidence of AI system stability under DoDI 5000.87. In operational AI systems—such as intelligence analysis, autonomous systems, or decision support tools—fewer false alarms mean faster, more reliable situational awareness and sustained compliance with Responsible AI mandates.

In the public sector, the innovation brings new capabilities to state and federal agencies managing high-volume, high-stakes AI applications like benefits eligibility and digital identity verification. By reducing false positives and improving anomaly resolution, agencies can prevent inappropriate denials, reduce the risk of bias, and meet NIST AI Risk Management Framework and HIPAA requirements. These protections were previously unavailable or prohibitively expensive using legacy methods.

In the commercial market, particularly for financial institutions governed by Federal Reserve SR 11-7 or upcoming OCC guidelines on explainable AI, AFFIRM's L1-enhanced monitoring introduces a differentiator: real-time, transparent anomaly scores that feed directly into compliance workflows. This offers a strategic edge over

conventional model governance platforms, reducing the rate of unnecessary investigations and helping identify novel fraud behaviors earlier.

Unlike point solutions that bolt drift detection onto black-box AI, Synergist's approach—rooted in L1-norm mathematical rigor and embedded in the operational AFFIRM platform—provides a complete, enterprise-ready solution. With seamless integration into SIEM systems, audit infrastructure, and existing compliance protocols, AFFIRM delivers trustworthy AI oversight at production scale.

In summary, this innovation introduces a new class of capability: statistically grounded, real-time AI anomaly detection with a 40% reduction in false positives, designed for the operational edge, but with broad applicability across civilian and commercial sectors. It is technically defensible, commercially viable, and positioned to become a category-defining solution for AI assurance.

Bahar Ghahremani bahar@newworldangels.com Mon Mar 16 10:11:31 2026

Biographies

Chris Pernicano, Research Collaborator (Synergist Technologies):

Mr. Pernicano is a seasoned technology leader with extensive C-level management experience in designing, deploying, and scaling cloud-based SaaS solutions across FinTech and HealthTech domains. In multiple ventures, he has guided business & technical strategy and negotiated high-value mergers and acquisitions. Through roles as CTO at Synergist Technologies and former Vice President of Engineering at Netsmart, he oversaw large teams executing enterprise software implementations, cybersecurity certifications, and AGILE development practices. He currently serves as Vice Chair for the North Broward Hospital District Board of Commissioners, where he oversees a $1.4B budget for critical healthcare services. Mr. Pernicano holds a BS in Mechanical Engineering, an MBA, and an MS in Engineering from the University of Florida.

Dr. Georgios Sklivanitis Research Collaborator (Florida Atlantic University):

Dr. Sklivanitis is a Schmidt Research Associate Professor and I-SENSE Fellow in the Department of Electrical Engineering and Computer Science at Florida Atlantic University. He is a Senior Member of the Center for Connected Autonomy and Artificial Intelligence (CA-AI) and a Faculty Fellow at the Institute for Sensing and Embedded Network Systems Engineering (I-SENSE). Dr. Sklivanitis received his Ph.D. in Electrical Engineering from the University at Buffalo, The State University of New York, and holds a five-year Diploma in Electronic and Computer Engineering from the Technical University of Crete, Greece. His research interests span autonomous radios, Ocean IoT, networked AI, and connected robotics. Through his work on robust signal processing techniques—including L1-norm subspace analysis—he has advanced the reliability and scalability of AI-enabled systems in complex environments.

References (Representative Prior Work):

[1] Tountas, K., Pados, D. A., & Medley, M. J.

Conformity Evaluation and L1-norm PCA of Tensor Data

This work introduces a conformity-driven approach to evaluating high-dimensional tensor data using L1-norm PCA. It demonstrates that L1-based metrics are better suited for identifying structural anomalies in noisy, real-world datasets. This concept of using residual conformity underpins the subspace learning and drift detection capabilities proposed in our AFFIRM enhancement.

[2] Colonnese, S., Scarano, G., Marra, M., Markopoulos, P. P., & Pados, D. A.

Joint Analysis and Segmentation of Time-Varying Data with Outliers

This paper establishes methods for decomposing time-series data using L1-subspace techniques that are robust to outlier bursts and distributional shift. Its treatment of temporal consistency and segmentation feeds directly into our methodology for modeling the evolving behavior of pre-trained AI systems under operational stress.

[3] Chamadia, S., & Pados, D. A.

Outlier Processing via L1-Principal Subspaces

This research proposes a novel algorithm for computing robust principal subspaces by assigning reliability weights and clustering them to remove anomalous data. The technique's demonstrated ability to reduce false alarms in contaminated datasets is directly aligned with our goal of reducing AI false positives by 40% in real-time monitoring scenarios.

[4] Varma, K., Nyman, L., Tountas, K., Sklivanitis, G., Nayak, A. R., & Pados, D. A.

Autonomous Plankton Classification from Reconstructed Holographic Imagery by L1-PCA-Assisted CNNs

In this application, L1-PCA filtering was used to improve CNN performance by preprocessing and filtering noisy or mislabeled holographic images. The study showed a 9% gain in classification accuracy (from 85% to 94%), which validates the foundational idea that L1-based preprocessing enhances downstream inference quality—an insight central to our anomaly detection strategy.

[5] Markopoulos, P. P., Karystinos, G. N., & Pados, D. A.

Optimal Algorithms for L1-Subspace Signal Processing

This work presents the first known exact algorithms for computing L1-norm principal components. These algorithms form the mathematical backbone of our proposed subspace generator and anomaly scoring engine. Without this foundational contribution, efficient and scalable deployment of L1-based anomaly detection in operational systems would not be feasible.